

Minimising total treatment time

In mathematics, a *computable real number* is one whose digits can be calculated to any desired precision by a terminating algorithm ¹. Here the code defines a_2 via a limit of sums. Noting that $\sum_{k=3}^{\infty} (1/2)^k = 1/4$ (since $1/2 + 1/4 + 1/8 + \dots = 1$ ²), one finds $2f(n) \rightarrow 1/2$ and $a_2 = 2f(n) + 1/2 + 0.25^n \rightarrow 1$ as $n \rightarrow \infty$. The terms 0.25^n and $72 \cdot 10^{-24}k$ are absolutely negligible, so a_2 is $1.000\dots$ with only an imperceptible excess. In effect $a_2 \approx 1$ (just slightly above 1), to any practical precision.

Optimization and solution

We must minimize $x_1 + x_2$ subject to the dose constraint $x_1 + a_2 x_2 = 1$ (with $x_1, x_2 \geq 0$). This is a linear program in two variables, so the optimum lies at a corner. The only feasible corners are $(x_1, x_2) = (1, 0)$ (all dose from therapy 1) or $(0, 1/a_2)$ (all dose from therapy 2). We compare:

- If $a_2 > 1$, then using only therapy 2 gives time $x_2 = 1/a_2 < 1$, which is faster than $x_1 = 1$.
- If $a_2 < 1$, then using only therapy 1 gives $x_1 = 1 < 1/a_2$, so therapy 1 only is faster.
- If $a_2 = 1$, any mix yields $x_1 + x_2 = 1$.

In our case a_2 is essentially > 1 , so the fastest option is $(x_1, x_2) = (0, 1/a_2)$. Numerically $1/a_2 \approx 1/(1 + \text{tiny}) \approx 1.00$. Thus to two-decimal accuracy the minimiser is

$$x_1 \approx 0.00, \quad x_2 \approx 1.00.$$

Answer: $x_1 = 0.00, x_2 = 1.00$ (to two-decimal accuracy, use only the faster second therapy). ² ¹

¹ Computable number - Wikipedia
https://en.wikipedia.org/wiki/Computable_number

² Geometric series - Wikipedia
https://en.wikipedia.org/wiki/Geometric_series